KETAN VILAS PATIL

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PROFESSIONAL SUMMARY

Systems-oriented engineering student with demonstrated experience delivering **production-grade AI, data-centric, cybersecurity, and full-stack solutions**. Proven capability across **end-to-end system design**—from data validation and backend services to frontend integration, security controls, documentation, and deployment. Strong foundation in **Python, applied machine learning, secure networking, and scalable system architecture**, with a consistent focus on **real-world impact, correctness, and reliability**.

EDUCATION

Bachelor of Technology – Computer Technology (Pursuing)
Dr. Babasaheb Ambedkar Technological University (DBATU), Lonere — **Sep 2024 – Present**
Diploma in Computer Engineering
Maharashtra State Board of Technical Education (MSBTE) — **Jun 2021 – Jun 2024**
HSC, Maharashtra State Board — **Feb 2020**
SSC, Maharashtra State Board — **Mar 2018**

TECHNICAL SKILLS

Languages: Python, JavaScript, C
AI / Data: Applied ML fundamentals, Pandas pipelines, data preprocessing, dataset validation
Backend: Flask, RESTful API design, rule-based inference engines, service-oriented architecture
Frontend: React (Hooks), component-based UI, responsive layouts, API integration
Security & Networking: Vulnerability assessment, penetration testing fundamentals, secure system design, CCNA-level networking
DevOps & Tools: Docker, Docker Compose, Git/GitHub, virtual environments
Databases: DBMS fundamentals, schema design, structured CSV datasets
Documentation: Technical documentation, integration guides, verification & QA reports

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PROFESSIONAL EXPERIENCE

Meta Inc. (Remote) — Jr. Network Analyst Intern — May 2023 – Aug 2023

- Optimized monitoring workflows handling ~15K daily requests, reducing latency by **12%**.
- Supported feature validation and testing, contributing to **20% faster** internal release cycles.

Sophos (Remote) — Jr. Security Analyst Intern — Sep 2023 – Dec 2023

- Analyzed firewall I/O across **200+ endpoints**, improving threat detection by **18%**.
- Tuned IDS configurations to reduce false positives by **10%**.

DPIIT, Government of India — Tech Support Intern (Telecommunication Dept.) — Jan 2024 – Jun 2024

- Supported communication systems for **1,000+ users**.
- Reduced average incident resolution time by **15 minutes** via systematic troubleshooting.

KEY PROJECTS

NAKSHATRAA – Precision Vedic Kundli Intelligence System — Feb 2026

Role: Full-Stack Engineer & AI System Integration
• **Architected** a production-ready Vedic astrology intelligence platform integrating **Swiss Ephemeris** celestial mechanics with deterministic prediction engines.
• Built a **bilingual (English/Marathi)** prediction layer across 3 life domains using structured astrological scoring models.
• Integrated **Rashi computation, Dasha timing overlays, and transit influence modeling** for temporally relevant insights.
• Engineered **asynchronous PDF generation and email pipelines (Celery + Redis)** with structured logging, health monitoring, and **rate limiting (5 req/min/IP)**.
• Designed a high-performance **glassmorphism UI** with adaptive motion systems and 40+ FPS rendering.
Impact: Delivered sub-second personalized life predictions; architected for multi-region scalability and multi-language expansion.
Tech: Python (Flask, Celery, Redis), JavaScript (Vite, Zustand), Swiss Ephemeris, Edge-TTS, Geopy, Docker Compose

SUNDAY – Offline-First Conversational Desktop AI Assistant — Feb 2026

Role: Full-Stack Engineer & Voice Systems Architect
• **Engineered** a fully offline, privacy-centric desktop AI assistant for Windows, macOS, and Linux.
• Architected an optimized voice pipeline: **wake-word → VAD → speaker verification → ASR → intent execution**, achieving <2.5s median latency.
• Implemented **owner-only speaker verification (ECAPA-TDNN)** with ≤1% false acceptance rate for secure OS command execution.
• Designed **context-aware planning intelligence** with episodic memory, ambiguity clarification, and unsaved-work protection.
• Built cross-platform **service integration (Windows services, systemd, launchd)** with automated voice enrollment workflows.
Impact: Delivered a deterministic, safety-first offline assistant capable of controlling complex desktop workflows via natural voice.
Tech: Python 3.11, Faster-Whisper, SpeechBrain, Silero VAD, openWakeWord, Coqui TTS, pytest, Docker Compose

AGRO VISION 2.0 – Verified Agricultural Remedies Intelligence System — Sep 2025

Role: AI / Data Engineering & System Integration

- Engineered a **production-ready remedies intelligence layer** extending an AI plant disease recognition system.
- Built a **region-validated database** covering **25+ diseases and 50 crops**, integrating **70+ CIBRC/ICAR-approved products** with 100% Maharashtra availability.
- Implemented **Pandas-optimized CSV pipelines** and authored **165+ pages of technical and user documentation** (developers, farmers, QA).
- Embedded **compliance-aware, safety-first constraints** (withholding periods, regulatory validation).
  Impact: Enabled actionable AI decision support with **₹10K–30K potential seasonal savings per farmer**; designed for multi-state scalability.
  Tech: Python, Pandas, AI system integration, data validation pipelines

Chikitsāmedhā – Full-Stack Medicine Interaction Checker — Nov 2025

Role : Full-Stack Developer

- Built a **complete full-stack pharmacological interaction analysis platform** for educational use.
- Implemented a **Flask backend** with **7 REST endpoints** and a **100-medicine dataset** spanning **30+ drug classes**.
- Developed a **rules-based inference engine (50+ rules)** covering CYP450 metabolism, additive toxicity, therapeutic duplication, and contraindications.
- Designed a **quantitative risk scoring algorithm (0–100)** and a **React frontend** with real-time search and visual risk stratification.
- Containerized with **Docker/Docker Compose** for reproducible, offline-capable deployment.
  Tech: Python, Flask, React, JavaScript, Docker, REST APIs

Penetration Testing – NMU Student Portal — 2024

Role : Security Analyst

- Remediated an **OTP verification bypass**; identified **5+ critical vulnerabilities** via MITM/session hijacking.
- Delivered a **20-page assessment**, reducing exposure by **~40%**.

ADDITIONAL TECHNICAL PROJECTS

• **Universal Image Tool (2025):** Batch image processing for **500+ images**, achieving **~70% size reduction**.
• **LC Issuer App (2025):** Automated **CSV-based certificate generation**, reducing human effort by **~70%**.
• **Secure Ops Dashboard (2025): Cybersecurity-inspired dashboard** with modular UI .

CERTIFICATIONS

- **CCNA (2024):** Networking fundamentals, routing & switching, network security.
- **AI for Cybersecurity Johns Hopkins University (Apr 2025):** AI-based threat detection and analysis.
- **IBM Cybersecurity Analyst (Jun 2025):** SIEM, incident response, penetration testing.
- **Meta Database Engineer (Sept 2025):** SQL, relational modeling, indexing, query optimization.
- **Meta Data Analyst (Oct 2025):** Data cleaning, EDA, analytics-driven decision making.
- **Google IT Support Professional (Nov 2025):** Networking, OS, system administration, IT security.
- **AWS Machine Learning (Nov 2025):** AWS ML services and cloud ML workflows.

ADDITIONAL INFORMATION

- Extensive experience producing **technical documentation, reports, and presentations**.
- **Languages:** English, Hindi, Marathi.

CAREER INTERESTS

Applied AI/ML Systems, **Backend & Full-Stack Engineering**, **Cloud-Native & Scalable Applications (AWS-focused)**, **Impact-driven domains: AgriTech, HealthTech, GovTech**.